

ServiceTitan Integration

What the AI can do

Summary

ServiceTitan Integration connects **ServiceTitan** (field service management platform for HVAC, plumbing, and electrical businesses) with **Newo Platform**.

It enables:

- Checking real-time dispatch capacity and presenting available time windows to callers
- Creating bookings that appear on the company's Calls screen for dispatcher action
- Looking up existing customers by phone number for returning callers
- Automatic LLM-powered matching of service requests to business units (HVAC, Plumbing, Electrical)
- Loading and caching business units and job types on session start
- Auto-configuring conversation canvas with service call booking and cancellation scenarios
- OAuth2 client credentials authentication with automatic token refresh on 401

This integration is designed for **home services businesses** (HVAC, plumbing, electrical contractors) who need **an AI receptionist to handle incoming calls, check availability, and create service bookings** through voice or chat.

Common use cases

- When a caller requests a service → Agent identifies the issue, checks dispatch capacity, and creates a booking
- When a caller asks about available times → Agent queries ServiceTitan dispatch capacity filtered by business unit
- When a returning customer calls → Agent looks up their profile by phone number
- When a caller wants to cancel or reschedule → Agent transfers to a human manager
- When a caller asks about job status → Agent collects details and transfers to team member

Features at a glance

Feature	Supported	Notes
Check Dispatch Availability	✓	By date range and business unit via Dispatch API
Create Booking	✓	With auto customer creation, appears on Calls screen
Customer Lookup	✓	Search by phone number via CRM API
LLM Service Classification	✓	Maps service request → business unit via Gemini Gen()
Static Data Sync	✓	Business units + job types loaded on session start
Canvas Auto-Setup	✓	Booking + cancellation + job status scenarios and intents
Multi-Business-Unit Support	✓	LLM resolves service type → correct business unit
Cancellation	✗	No API for booking cancellation — transfers to manager

Reschedule	✗	No direct reschedule — transfers to manager
Job Status Lookup	✗	Intent defined, but transfers to manager (API not wired)
Payment Processing	✗	Not implemented

Before You Start

Before installation:

- Active **ServiceTitan** account with API access enabled
- **Client ID** and **Client Secret** from the ServiceTitan Developer Portal (environment-specific)
- **App Key** (ST-App-Key) from the ServiceTitan My Apps portal (environment-agnostic)
- **Tenant ID** from your ServiceTitan account
- At least one **Business Unit** configured in ServiceTitan (e.g., HVAC, Plumbing, Electrical)

Setup

Create the Newo project

If this project already exists in **Newo Builder**, skip this subsection and continue with the integration-specific settings below.

1. In **Newo Builder**, open the projects list and click **Create Project** (or **Create New Project** from the top-right menu).



Create New Project menu in Newo Builder

2. Fill **IDN** and **Title**. The exact names do not matter; use any clear names your team will recognize.
3. In **Registry**, choose the release channel:
 - **staging** — the newest module fixes appear here first. Use it when you need the latest fix, but expect possible unfinished changes.
 - **production** — the final stable version for live projects.
4. In **Module**, select the module for this integration.



Create Project form showing IDN, Title, Registry, and Module fields 5. Leave **Module version** on **Latest**

version unless support tells you to pin a specific version, then click **Create**.

3.1 Step 1 — Generate Credentials in ServiceTitan

1. Log in to the **ServiceTitan Developer Portal** (<https://developer.servicetitan.io>)
2. Navigate to **My Apps** → create or select your application
3. Copy the **App Key** (this is environment-agnostic)
4. Go to your application's **Credentials** section
5. Select the environment (Production or Integration/Sandbox)
6. Copy the **Client ID** and **Client Secret**
7. Note your **Tenant ID** from the ServiceTitan account settings

3.2 Step 2 — Connect in Platform

1. Open **Newo** → **Projects**
2. Set the following attributes in **Builder / Attributes**:

Attribute	Required	Default	
servicetitan_client_id	✓	—	C
servicetitan_client_secret	✓	—	C s
servicetitan_app_key	✓	—	A a
servicetitan_tenant_id	✓	—	T
servicetitan_base_url	✗	https://api.servicetitan.io	A i
servicetitan_auth_url	✗	https://auth.servicetitan.io/connect/token	C h i
servicetitan_booking_source	✗	AI Receptionist	S
servicetitan_enable_slot_check	✗	True	E
servicetitan_enable_booking	✗	True	E
servicetitan_setup_scenarios	✗	True	A t
servicetitan_override_agent_attributes	✗	True	C
servicetitan_default_business_unit_id	✗	—	F r

3. Click **Save + Publish All**

4. On publish, the SetupFlow automatically:

- Prepares the connections for ServiceTitan API and token endpoint
- Sets all customer attributes with metadata
- Registers booking and availability tools on ConvoAgent
- Injects availability and booking payload schemas

5. On first conversation, StaticDataFlow loads business units and job types from ServiceTitan

How to use the integration

4.1 How the Integration Works

The integration uses ServiceTitan **Dispatch API** for availability, **CRM API** for customers and bookings, and **Settings/JPM APIs** for reference data. Service requests are classified to the correct business unit via LLM.

- A **Trigger** is a system event that starts a flow
- An **Action** is the API operation performed in ServiceTitan

Canvas scenarios

When `servicetitan_setup_scenarios` is `True` (default), Publish All automatically adds:

Scenarios:

- **"Scheduling Service Call via Agent"** (`servicetitan_booking_scenario`) — 8-step guided booking flow with code-phrases for availability check and booking creation

Intent Types:

- **"[L] Service Call Request via Agent"** (`servicetitan_booking_intent`) — caller wants to schedule service
- **"[T] Job Status Inquiry"** (`servicetitan_job_status_intent`) — caller asking about existing job
- **"[T] Cancellation or Reschedule Request"** (`servicetitan_cancellation_intent`) — caller wants to cancel/reschedule → transfer to manager

These can be customized on the canvas after setup — modified scenarios are not overwritten.

Where to test

Testing can be done through the **Newo conversation interface** (chat or voice). A mock server is available at `mocks/servicetitan/routes.yaml` for testing without real API credentials.

How to test that everything works

To test the integration:

1. **Setup:** Publish the project and verify connectors are created
2. **Static Data:** Start a conversation — verify `servicetitan_business_units` and `servicetitan_job_types` are populated
3. **Availability:** Ask about a service — verify dispatch capacity returned with time windows
4. **Booking:** Complete a booking — verify booking created in ServiceTitan Calls screen
5. **Cancellation:** Ask to cancel — verify agent transfers to manager

If no action occurs:

- Ensure credentials are set (`servicetitan_client_id` , `servicetitan_client_secret` , `servicetitan_app_key`)
- Verify `servicetitan_tenant_id` is correct
- Confirm `servicetitan_base_url` points to correct environment (production vs sandbox)
- Check feature flags (`servicetitan_enable_slot_check` , `servicetitan_enable_booking`)
- Re-publish after configuration changes
- Verify `servicetitan_business_units` is populated (check logs for StaticDataFlow)

Note: This integration creates real bookings in ServiceTitan. Bookings appear on the Calls screen and are visible to dispatchers.

FAQ

Q: What is the difference between a Booking and a Job in ServiceTitan? A: A **Booking** is an incoming inquiry/request that appears on the Calls screen. A **Job** is a confirmed work order on the dispatch board, created by a dispatcher from a booking. The AI receptionist creates Bookings, not Jobs directly.

Q: How does the agent determine which business unit to query for availability? A: The `_resolveBusinessUnitSkill` uses Gemini LLM (Gen, temperature=0.1) to classify the caller's service request against the loaded list of business units and job types. For example, "my AC isn't cooling" → HVAC. If classification fails, it falls back to `servicetitan_default_business_unit_id`.

Q: Why can't the agent cancel bookings? A: ServiceTitan's API does not provide an endpoint for cancelling bookings. Cancellation/rescheduling is handled by human dispatchers. The agent recognizes cancellation requests and transfers the caller to a manager.

Q: How are tokens managed? A: OAuth2 client credentials flow. Tokens are valid for 15 minutes with no refresh token. On 401 responses, UtilsFlow automatically re-authenticates and retries (up to 3 times). Tokens are cached in `servicetitan_access_token` customer attribute.

Q: When is static data (business units, job types) loaded? A: On every `session_started` event (start of each conversation). Data is cached — if already loaded, the API call is skipped. To force reload, clear `servicetitan_business_units` and `servicetitan_job_types` attributes.

Q: Can I connect to the ServiceTitan Sandbox? A: Yes. Set `servicetitan_base_url` to `https://api-integration.servicetitan.io` and `servicetitan_auth_url` to `https://auth-integration.servicetitan.io/connect/token`. Use sandbox Client ID and Client Secret.

Q: What customer information is needed for booking? A: The caller's phone number is required for customer lookup/creation. Full name is used when creating a new customer. The booking summary is generated from the conversation context describing the service needed.